

FORGE STAR

Technical and investor brief

Investors · Counsel · Systems & RF · Software · Hardware · Due diligence

MYFORGE PTY LTD · FORGE ECOSYSTEM

Brief · November 2026 · Confidential

00 · HOW TO READ THIS

One document, several readers

FOR: EVERYONE

FORGE STAR is a South African company building a locally owned low-Earth-orbit network for IoT, maritime AIS and sovereign data relay. It runs as a ground and services company first (revenue before orbit), then becomes a satellite operator. Each section below is written for a specific reader, and you can read any of them on its own.

Section	Written for	What you get
01 Executive summary	Investors, board	The case in one page
02 Market and business model	Investors	Customers, revenue lines, why now
03 Financials and stress tests	Investors, CFO	Capital need, revenue path, downside cases
04 Regulation and ownership	Counsel, investors	ICASA, ITU, the 30% HDG rule
05 System architecture	Everyone	How the network works end to end
06 Constellation and RF design	Systems and RF engineers	Orbits, coverage, delivery time, link budget, coding
07 Flight software	Software and flight engineers	F´ deployment, security, safe mode
08 Ground segment and platform	Software, operations, customers' IT	Gateway, data platform, API, billing
09 Hardware: build, buy, bridge	Hardware and RF engineers, partners	What we build, what we buy, who covers the gaps
10 Evidence and maturity	Technical due diligence	Tests, CI, TRL, what is verified
11 Path to launch	Everyone	Timeline, critical path, next 90 days
12 Risks	Investors, board	What could break the plan
13 Sources and caveats	Everyone	What was checked, and what still needs confirming

01 · EXECUTIVE SUMMARY

FORGE STAR in one page

FOR: INVESTORS AND BOARD

FORGE STAR is Africa's sovereign satellite data network: locally owned LEO infrastructure that connects the assets, vessels and people that terrestrial networks will never reach, with the data landed and hosted in South Africa.

Question	Answer
What we sell	Satellite IoT connectivity, maritime AIS data, sovereign data relay; later wholesale capacity to mobile operators and constellation-in-a-box for African states
Who buys	Mining, agriculture, water, conservation, fisheries and Navy, logistics corridors, the SA government, FORGE ecosystem ventures
Why us	Compliant with ICASA's local-ownership rules where Starlink is not licensed; sovereign data hosting; Africa-first design
Stage 3 design	48 satellites (42 at 45° + 6 at 10°) and 6 gateways: 95% of messages delivered within 26 minutes at every modelled customer site
Measured, not just modelled	The full platform runs end to end on a simulated constellation: 95% of satellite messages delivered within 9 minutes in a live run
Capital	~$50M equity to an operational network (Stage 3); gross need ~$47M (base case)
Revenue	~$37.5M a year at Stage 3 (year 6, base case)
First payload in orbit	~April 2029 on the fast track (hosted payload); own satellite ~June 2029
What it waits on	Spectrum counsel and ICASA, the bus partner, the lab grant, signed LOIs — not software

What already exists. The flight software (NASA F´), the store-and-forward core in C, the radio link layer, the gateway, the data platform, dashboards, customer API, billing, the constellation and financial models, an in-house PCB pipeline and a readiness tracker. All of it is in one repository and tested on every change.

02 · MARKET AND BUSINESS MODEL

Who pays, for what, and why now

FOR: INVESTORS

Why now

- **Starlink is not licensed in South Africa.** ICASA's 30% historically-disadvantaged-group (HDG) ownership rule applies; FORGE STAR is designed to comply.
- **The equity-equivalent route is contested.** ICASA says it needs a change to the Electronic Communications Act. This is the main threat to the moat, and the plan does not depend on Starlink staying out.
- **Chinese constellations are behind target and carry alignment risk.** Other operators do not serve African sovereign demand or mass IoT.
- **Early costs have collapsed.** Open-source flight and ground software and AI-assisted design make a demonstrator affordable.

Customers and pricing

Segment	Buyers	Offer	Pricing
Maritime and security	SA Navy, DFFE, SAMSA, ports, insurers	AIS vessel tracking, illegal-fishing detection	Annual subscription or contract
Mining and energy	Mining houses, Eskom, IPPs	Remote asset monitoring	Per device per month + platform fee
Agriculture and water	Farms, co-ops, water boards	Sensors beyond cell coverage	Per device per month
Conservation	SANParks, reserves	Collars, fence sensors, ranger messaging	Per device or per reserve
FORGE ecosystem	Freight, USV, VOCAL, ForgePay, EYE	Tracking, links, offline messaging	Internal contracts, first references
SA government	DCDT / SA Connect, provinces	Sovereign capacity, rural sites	Multi-year contract, part prepaid
Mobile operators (later)	MTN, Vodacom, others	Wholesale NB-IoT satellite capacity	Wholesale + revenue share

Pricing reality check. New LEO IoT entrants now price at about $1 per device per month (Myriota from $0.99; Sateliot targets about €1). FORGE STAR competes on sovereignty, local support and data landing in SA, not on undercutting. IoT is a minority of Stage 3 revenue; the **SA government capacity contract** is the line the plan depends on most.

03 · FINANCIALS AND STRESS TESTS

Capital, revenue path and downside cases

FOR: INVESTORS AND CFO

Generated from `model/` (years 1–6, Stage 0 to Stage 3). Illustrative; not financial advice.

Year	Satellites	Revenue (base)	Revenue (low)	Revenue (high)
1	0	$0.3M	$0.2M	$0.4M
2	1	$1.4M	$0.8M	$1.8M
3	2	$3.9M	$2.3M	$5.1M
4	10	$16.5M	$9.9M	$21.5M
5	30	$29.4M	$17.6M	$38.2M
6	48	$37.5M	$22.5M	$48.8M

Stress test	Year-6 revenue	Year-6 EBITDA	Read
Base	$37.5M	$12.6M	No month below zero cash
IoT at ~$1/device	$31.6M	$9.2M	Still in the Stage 3 range
No SA government contract	$19.5M	$2.2M	Below the Stage 3 range: the biggest commercial risk
Low case (slow sales, higher costs)	$22.5M	$3.0M	Cash short ~$1.5M in year 3: needs a larger seed or earlier Series A

Funding plan

Round	When	Amount	Purpose
Founder, bootstrap, grants	Year 1	~$0.55M	Stage 0 platform, lab, licences
Seed (on signed LOIs)	By month 12	~$2.5M	Hosted payload, own demonstrator start
Series A	Year 3	~$22M	Early service (6–12 satellites)
Series B + government prepayment	Year 5	~$25M + prepayments	Operational network (48 satellites)

Equity to Stage 3: ~$50M. **Gross capital to Stage 3:** ~$47M base, ~$60M in the low case. The seed must close by month 12; the readiness plan shows that is only reachable by raising on signed LOIs rather than waiting for paying customers.

04 · REGULATION AND OWNERSHIP

Permission to transmit and the 30% rule

FOR: COUNSEL AND INVESTORS

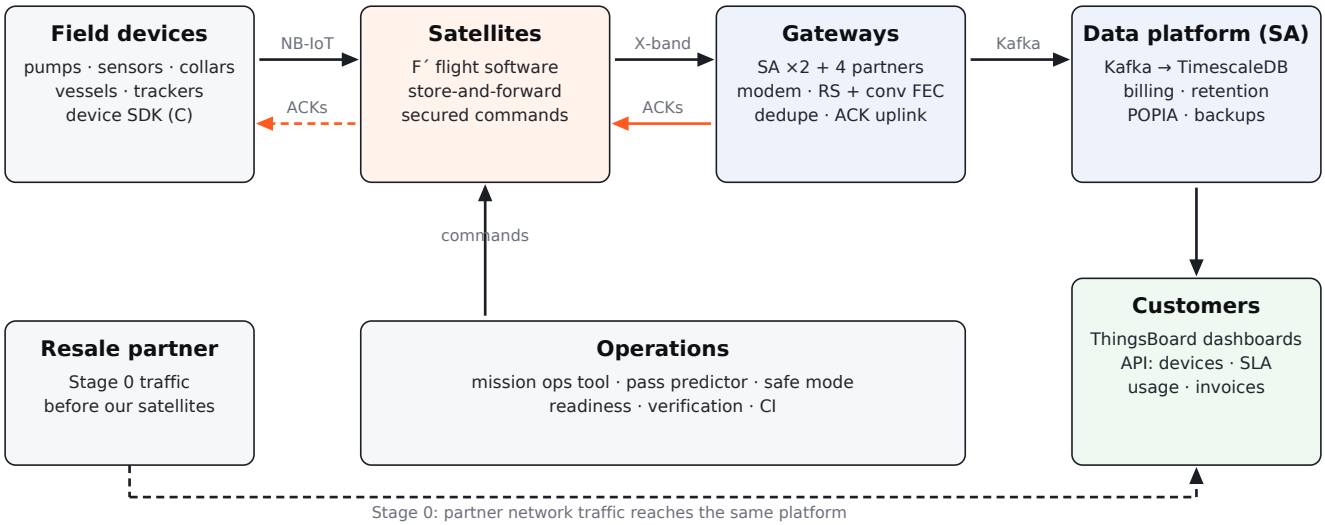
Item	Position	Status
ICASA licensing	ECNS/ECS + radio frequency spectrum licence; test licence for the lab	Not started: counsel to be briefed
ITU filing	Modest NGSO filing through ICASA; draft parameters generated (`docs/itu/FILING_PARAMETERS.md`)	Draft ready for counsel
Bands (draft)	NB-IoT NTN n256 (1980–2010 / 2170–2200 MHz), AIS receive (VHF), X-band feeder (8025–8400 MHz)	To confirm against RR and ICASA band plan
Deployment milestones	Res. 35: 10% / 50% / 100% within 2 / 5 / 7 years of the bring-into-use deadline	Filing 96 satellites means 95 needed by the final milestone; file closer to 48 if growth is uncertain
30% HDG ownership	Cap table keeps HDG ownership above 50% through Series B, if each round includes HDG-qualifying investors	Founder flags are placeholders: confirm with counsel
POPIA	Data hosted in SA; 24-month default retention; audit log; Information Officer to appoint	Checklist drafted

After round	Post-money	HDG share
Founding	—	75.0%
Seed	$12.5M	66.0%
Series A	$82M	54.8%
Series B	$245M	52.2%

05 · SYSTEM ARCHITECTURE

How the network works end to end

FOR: EVERYONE



Orange: acknowledgements (ground → satellites → devices, so devices stop re-sending). Dashed: Stage 0 resale traffic, before our satellites fly.

How a message travels.

- A device sends a reading over NB-IoT NTN. Every FORGE STAR satellite overhead takes a copy and holds it on board.
- When a satellite passes a gateway, it downlinks its stored messages in error-corrected link blocks.
- The gateway decodes them, removes duplicate copies, stores each message once and sends acknowledgements back up.
- Satellites relay the acknowledgements: devices stop re-sending, satellites drop copies they no longer need.
- Messages flow through Kafka into TimescaleDB in South Africa; customers see them in dashboards and through the API, and are billed monthly.

Stage 0 uses the same platform with traffic from a resale partner, so customers sign up and pay before our satellites fly and move across with no change on their side.

Layer	What it is	Status
Devices	NB-IoT NTN modules + FORGE STAR device SDK (C)	SDK built and tested; module integration needs hardware
Space segment	6U–12U satellites on a bought bus; our payload and flight software	Software built; payload hardware to build
Gateways	SA-owned + leased partner stations; modem, FEC, dedupe, acks	Software built and tested
Data platform	Kafka, TimescaleDB, ThingsBoard, API, billing	Built; runs end to end; production overlay ready
Operations	Mission ops tool, procedures, readiness and verification tracking	Built

06 · CONSTELLATION AND RF DESIGN

Orbits, delivery time, link budget and coding

FOR: SYSTEMS AND RF ENGINEERS

Stage 3 constellation

Shell	Satellites	Planes	Inclination	Altitude
A (main)	42	7	45°	550 km
B (equatorial)	6	1	10°	550 km

Gateways: Hartebeesthoek, Cape Town (owned), Walvis Bay, Mauritius, Malindi, Lagos (leased, data relayed to SA). Orbit model: circular + J2 secular drift, checked against SGP4 (planes agree within 3 km over 3 days).

Delivery time (95th percentile)

Configuration	Johannesburg	Maputo corridor	Lusaka	Dar es Salaam	Nairobi	Lagos
24 sats, SA gateways, 22° mask	1.6 h	81 min	1.8 h	3.4 h	5.8 h	4.3 h
36 sats, 4 gateways, 16° mask	44 min	71 min	12 min	44 min	1.8 h	1.9 h
Stage 3 design (42+6, 6 gateways)	8 min	8 min	26 min	11 min	9 min	11 min

Live platform run on the Stage 3 design: **95% of satellite messages within 9 minutes** (maximum 14 minutes). Two design rules came out of this work: devices must **re-send until acknowledged** (single-copy hand-off left messages stranded on satellites heading away from gateways), and **gateways are the cheap lever** for sites far from SA.

Link budget (illustrative parameters, `forgestar/params/rf.yaml`)

Link	Closes (3 dB margin) from	Note
NB-IoT uplink (23 dBm device → 14 dBi patch)	~22°	Sets the baseline mask
NB-IoT downlink (2 W, ±30° patch)	~37°	The payload design driver
AIS receive (VHF monopole)	horizon	Collision-limited, not budget-limited
Feeder downlink (X-band → 3.7 m dish)	~4°	Comfortable

Antenna trade

Option	Usable mask	DC power	Est. cost
Patch 14 dBi, 2 W (baseline)	37°	12 W	~$15k
Patch + deeper NB-IoT repetition	26°	12 W	~$15k
Isoflux 7 dBi, 5 W	21°	25 W	~$45k

Option	Usable mask	DC power	Est. cost
Isoflux 5 dBi, 8 W	16°	38 W	~$55k
4×4 phased array	20°	30 W	~$150k

Demonstrator: patch + deeper repetition. Stage 2 onward: shaped isoflux antenna, provided the bus supplies ~25–40 W during passes.

Link layer

Element	Choice	Verified
Outer code	Reed–Solomon (255,223), CCSDS parameters, dual basis (libfec table)	Own encoder and decoder; corrects up to 16 bytes per codeword
Inner code	CCSDS convolutional K=7 r=1/2, soft-decision Viterbi	Decodes at Eb/N0 ≈ 2.5 dB (RS alone needs ~6 dB)
Randomiser, sync	CCSDS pseudo-random sequence; 0x1ACFFC1D marker	Matches CCSDS 131.0-B sequence
Modulation	BPSK, root-raised-cosine; receiver handles Doppler, timing, phase	Software modem tested with offsets and noise

07 · FLIGHT SOFTWARE

F´ deployment, security and fault handling

FOR: SOFTWARE AND FLIGHT ENGINEERS

The payload software is a NASA F´ (v3.5.1) deployment with a portable C99 core (`flight/libforge`, no dynamic memory) that is byte-identical to the ground implementation (shared test vectors).

Component	Responsibility
StoreForward	Onboard store (16,384 messages), frame building, link blocks; ACK handling; safe-mode inhibit; persistence across reboots; memory scrubbing
CmdAuth	Every uplinked command ChaCha20-encrypted and HMAC-SHA256-authenticated with a monotonic counter; forged, corrupted or replayed commands are dropped before the dispatcher
ModeManager	SAFE mode on low bus voltage, lost bus reports or operator command; return to NOMINAL needs a command and healthy power
PassPredictor	Onboard gateway pass prediction from uplinked orbit elements (matches the ground model to 1e-9 km)
PayloadSim	Hardware stand-in (NB-IoT messages, bus power, downlink radio) for running without hardware; replaced by real drivers
Standard F´	Rate groups, command dispatcher, event logger, telemetry, parameters, buffer manager, framer, TCP ground link

Property	Evidence
Runs on target-class hardware	Cross-compiled for 64-bit ARM and run under QEMU on every CI build
Nothing lost or reordered	Downlink decoded on the ground: 720 messages, in order, every block decoded
Survives reboot	Store restored after restart (integration test on the real binary)
Security	Unsigned, replayed and tampered commands rejected; opcode not visible on the wire
Radiation upsets	Corrupted stored records detected by per-record CRC and discarded, never downlinked

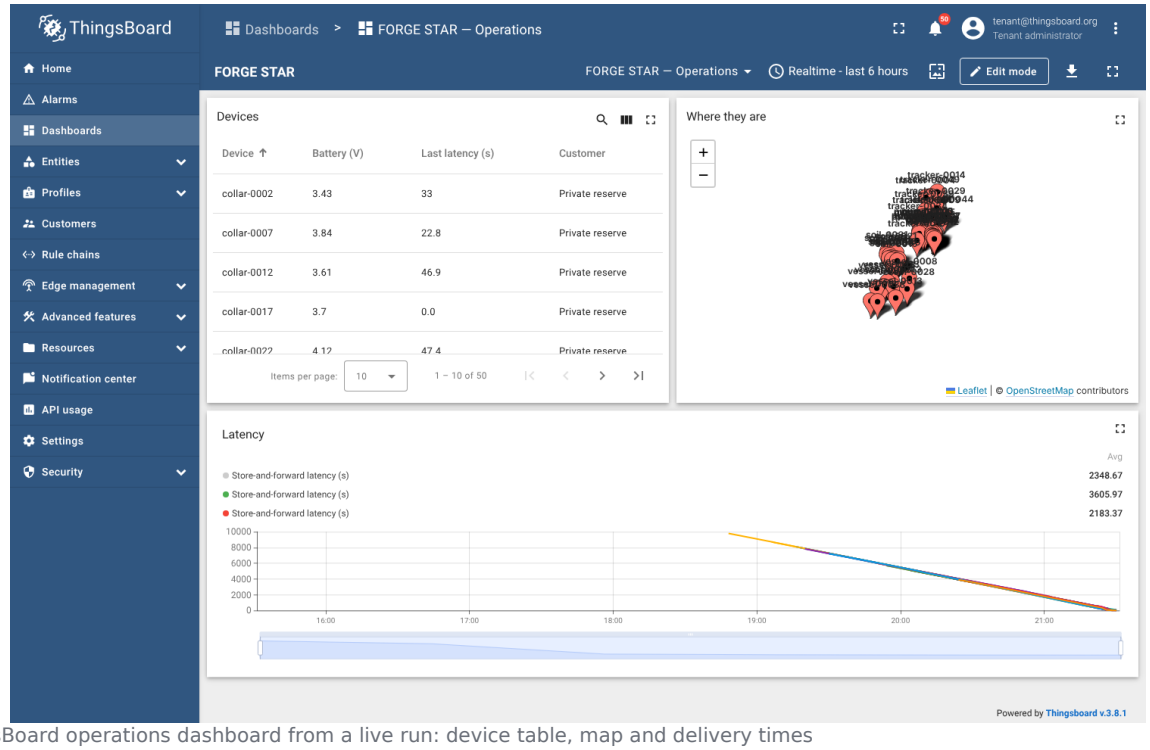
Still to build (needs hardware): the NB-IoT NTN base station on the payload SDR (OpenAirInterface NTN), real radio and bus drivers, the payload computer port, hardware-in-the-loop tests.

08 · GROUND SEGMENT AND PLATFORM

Gateway, data platform, API and billing

FOR: SOFTWARE, OPERATIONS AND CUSTOMERS' IT

Service	What it does
Gateway	Reads IQ recordings, demodulated bytes or frames (Kafka, files, SatNOGS Network); RS + Viterbi decoding; duplicate removal; ACK uplink
Ingest	Kafka → TimescaleDB; ThingsBoard forwarding
Customer API	Per-customer keys; devices, messages, latest reading, SLA, usage, invoices; rate limits, audit log, metrics, key revocation
Billing	Monthly invoices per active device, minimum charge, SLA credit when the 95% delivery target is missed
Dashboards	ThingsBoard: devices, map, delivery-time chart (provisioned by script)
Mission operations	Signed, encrypted commanding and telemetry decoding for the flight software; procedures for passes, safe mode, key rotation
Production overlay	TLS (Caddy), nightly backups, no database or Kafka ports on the host, required secrets, 24-month retention



ThingsBoard operations dashboard from a live run: device table, map and delivery times

09 · HARDWARE: BUILD, BUY, BRIDGE

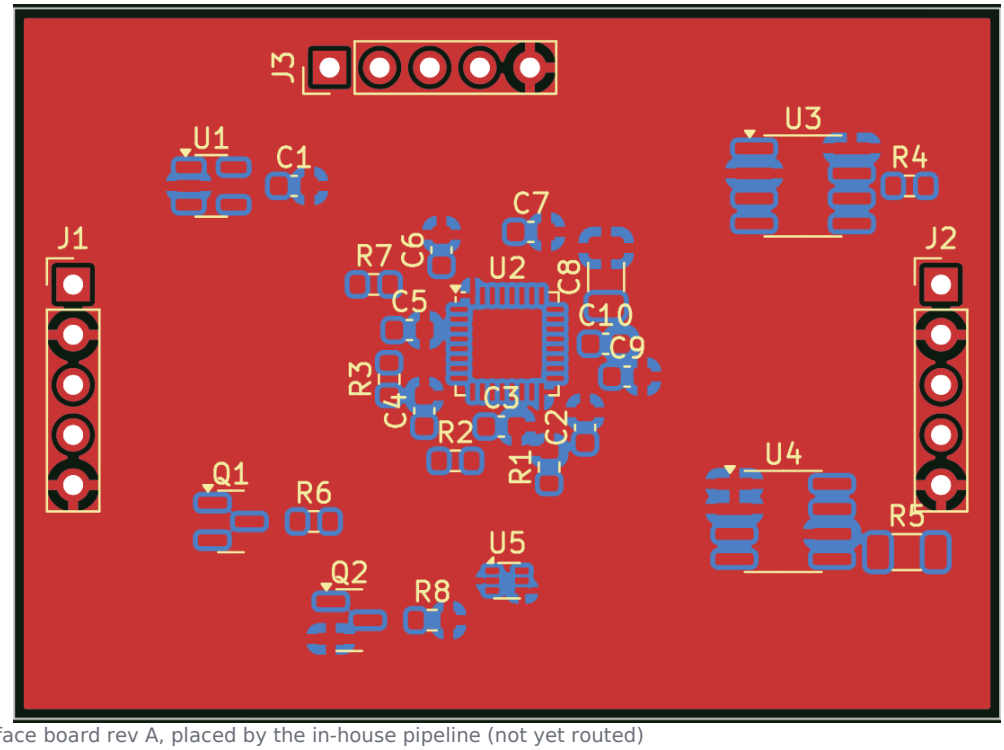
What we design, what we buy, and who covers the gaps

FOR: HARDWARE AND RF ENGINEERS, PARTNERS

Boards: heypcb.ai plus our pipeline

heypcb.ai designs the circuit boards (schematics, routing, Gerbers, BOMs, KiCad projects), starting from its library of open-hardware boards. Every board comes back as a KiCad project, so our in-house pipeline (SKiDL → KiCad → design rule checks → fab files) checks and packages it on every change.

Board	Designed by	Status
Payload interface board (STM32, CAN, radio power switch, current and temperature)	heypcb.ai from our netlist; checked by our pipeline	Schematic and placement done; routing next
Payload processor carrier	heypcb.ai (module bought)	Module to choose
S-band radio carrier (digital, power, clock)	heypcb.ai; RF layout by the RF engineer	Not started
Lab test boards	heypcb.ai	Not started



Payload interface board rev A, placed by the in-house pipeline (not yet routed)

The gaps heypcb.ai does not cover, and how we bridge them

Gap	Bridge	Notes from the research (to confirm)
Bus integration, structure, thermal	Hosted-payload platform (e.g. Loft Orbital Payload Hub) or the bus partner's integration service (Dragonfly, CubeSpace)	Loft integrates payloads on a standard hub (reported ~2 days); no public pricing. Dragonfly offers 50–150 kg-class buses; no public hosted-payload offer

Gap	Bridge	Notes from the research (to confirm)
High-performance RF	Flight-heritage radios: EnduroSat, Alén Space, IQ Spacecom, Syrlinks	EnduroSat list prices: S-band transceiver ~$13–14k; X-band transmitter ~$32k. A TT&C transceiver is not an NB-IoT base station: the payload still needs an SDR
Antennas	Antenna Magus (design library), openEMS / CST / HFSS (simulation), range test	Antenna Magus Learning Edition is free (12 antenna types); commercial licence quote-only
Radiation	SPENVIS analysis, heritage parts, proton testing at iThemba LABS (Cape Town)	SPENVIS is free. iThemba LABS has a proton single-event-effect facility (first in Africa); external access terms to confirm
Structural and thermal analysis	SimScale, FreeCAD + CalculiX; bus partner's thermal model; vibration and thermal-vacuum tests	SimScale has a free tier (public projects only)
Systems engineering	Capella (open source) for interfaces and budgets	Valispace: no startup programme found; reported acquired by Altium

Bought in

Item	From	Illustrative cost
Satellite bus	Dragonfly Aerospace or CubeSpace	Part of ~$1.8M for the first own satellite
Hosted-payload slot	Bus partner or a hosting platform	~$350k
Launch	Rideshare broker	~$250k + ~8% insurance
NB-IoT NTN device kits	Module vendors	$100–300 each
Lab kit	SDRs, GPS clock, attenuators, shielded box, analyser	~$12–25k (TIA/DSI grant)

People still needed: an RF engineer, and a part-time systems or mechanical engineer to own interfaces and tests.

10 · EVIDENCE AND MATURITY

What is tested, and how mature each part is

FOR: TECHNICAL DUE DILIGENCE

Evidence	Count
Python tests	66 (includes live tests against the flight binary)
C test suites	4 (core, command security, orbit, device SDK)
F´ unit tests	6
CI jobs on every push	4: Python and analysis, platform images and production overlay, flight software (native and ARM), hardware pipeline
Cross-language checks	C and Python produce identical frames, link blocks, convolutional code, ACKs and orbits

Subsystem	TRL now	Needed for launch
Store-and-forward core (C)	4	6
Flight software (F´)	4	6
Radio link layer	4	6
Device SDK	4	5
Ground gateway	4	6
Data platform	5	7
Constellation design	4	5
Payload interface board	3	6
NB-IoT NTN payload radio	2	6
Payload antenna	2	6
Satellite bus	bought	flight-proven

Verification register (`docs/verification/VERIFY.md`): 16 claims tracked, 12 web-checked, 0 yet confirmed by counsel or quotes; 34 of 36 radio parameters still illustrative pending datasheets.

11 · PATH TO LAUNCH

Timeline, critical path and the next 90 days

FOR: EVERYONE

Milestone	Fast track	Current plan
ICASA pre-application meeting	Apr 2027	Apr 2027
Lab grant awarded	May 2027	May 2027
ICASA test licence	Aug 2027	Aug 2027
Seed round closed	Nov 2027	Apr 2028
Lab proves NB-IoT through the satellite channel	Nov 2027	Nov 2027
Payload frozen against the bus/host interface	Mar 2028	Mar 2028
ICASA spectrum licence	Jun 2028	Jun 2028
Hosted payload in orbit	**Apr 2029**	Jul 2029
Own 6U–12U satellite in orbit	**Jun 2029**	Jul 2030
Early service (6–12 satellites)	Jul 2030	Jan 2032

Fast track = seed raised on signed LOIs, own demonstrator built in parallel with the hosted payload, Series A on the hosted-payload result.

The critical path runs through spectrum: counsel → ICASA pre-application meeting → test licence → lab → antenna decision → payload freeze → hosted-payload slot → launch.

Next 90 days

- **Spectrum counsel:** brief with the ITU parameter sheet; request the ICASA meeting and a lab test licence; decide the filing size.
- **Bus partner and host RFQs:** Dragonfly and CubeSpace, plus one hosting platform; ask for 25–40 W payload power during passes and the interface control document.
- **Lab grant:** TIA/DSI application with the 12-week test plan and bill of materials.
- **Payload interface board:** route with heypcb.ai, fab 5 boards, bring up with the flight core.
- **RF quotes:** S-band SDR and power amplifier, X-band transmitter, S-band patch.
- **Radiation:** SPENVIS analysis for our orbits; contact iThemba LABS about access.
- **Revenue:** signed LOIs from FORGE Freight, the maritime venture and one mining group; first SatNOGS station live.

What we are asking for

From	Ask
Investors	Seed of ~$2.5M by month 12, with an HDG-qualifying anchor
Grant bodies	Lab funding for the NB-IoT NTN test-bed
Partners	Bus or hosting quotes and interface documents
Customers	Letters of intent and pilots on the Stage 0 platform

12 · RISKS

What could break the plan

FOR: INVESTORS AND BOARD

Risk	Why it matters	Mitigation
Spectrum (ICASA/ITU)	No licence, no service; milestones can truncate a filing	Early counsel; modest filing; draft parameters ready
Ownership rules change	Equity equivalents could let Starlink in	Sell sovereignty and IoT, not broadband
Sovereign contract	Without it, year-6 revenue falls to ~$19.5M	Start the government conversation early; prepayment terms
IoT price pressure	Entrants at ~$1/device	Compete on sovereignty, local service, SLA
Payload RF	Downlink closes only from ~37° on a simple patch	Isoflux antenna; deeper repetition for the demo; RF hire
NB-IoT NTN software maturity	Open-source base station is the biggest unknown	Lab test-bed first; vendor fallback
Capital timing	Seed needed by month 12	Raise on signed LOIs; grants as bridge
Launch slippage	Rideshare dates move	Hosted payload first; schedule buffers

13 · SOURCES AND CAVEATS

What was checked, and what still needs confirming

FOR: EVERYONE

Product names, programmes and facility details come partly from general knowledge and partly from a web check on 2–3 November 2026, not from vendor quotes. Vendors' offers, pricing and availability change, so every figure in sections 02, 04 and 09 needs confirming directly before commitments. Each item is tracked in the verification register (`docs/verification/VERIFY.md`, items R1–R4, M1–M2, C1–C2, O1, V1–V7) and the readiness plan (items H1–H7).

Item	Checked	Still to confirm	Owner
Starlink / ICASA status	Web-checked	Counsel opinion	Spectrum counsel
ITU Resolution 35 milestones	Web-checked	Consolidated Radio Regulations text	Spectrum counsel
IoT pricing	Web-checked (vendor marketing)	Competitor quotes	Sales lead
Starlink scale	Web-checked (S-1 via secondary sources)	Primary filing	Founder
Hosting platforms	Web-checked; no public prices	Quotes from Loft and one other	Founder
Bus partners	Web-checked; no public hosted offer	RFQ responses	Founder
RF hardware prices	List prices (satsearch, vendor pages)	Formal quotes	RF engineer
Antenna Magus, SimScale, SPENVIS	Web-checked	Licence terms for commercial use	Systems engineer
iThemba LABS radiation testing	2020 thesis	Current access and pricing	Systems engineer
Radio parameters	Illustrative	Datasheets and measurements	RF engineer
HDG ownership flags	Placeholders	Verification	Counsel

Key sources

- ICASA and Starlink: advanced-television.com (May 2026); furtherafrica.com (Aug 2026); newsroom.co.za; techinafrica.com
- ITU NGSO milestones: itu.int (Resolution 35 announcement; 2024 workshop on Resolutions 8 and 35)
- IoT pricing: itbrief.com.au (Myriota); myriota.com; iotone.com (Sateliot); hubble.com (NTN cost breakdown)
- Starlink scale: driveteslacanada.ca; kalkine.com; earthsims.com
- Hosting and buses: loftorbital.com; payloadspace.com; satsearch.co (Dragonfly); spaceinafrica.com (Transporter-17)
- RF hardware: satsearch.co (EnduroSat S-band, X-band, SDR)

- Tools and testing: 3ds.com (Antenna Magus); spenvis.oma.be; pricingsaas.com (SimScale); scholar.sun.ac.za (iThemba LABS proton SEE facility); elreferente.es (Valispace)

Full links are in `docs/verification/register.yaml`. Not financial or legal advice.